

Albert J. Lee

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RESEARCH SUMMARY

ML Ph.D. candidate working on **reliable LLM reasoning in clinical decision tasks**: whether a model applies the rules and evidence in front of it correctly, and how to fix it when it does not. I design **counterfactual evaluations** that separate correct rule application from irrelevant sensitivity (e.g., treatment eligibility that should change when the rules change but not when only the patient's stated goal changes), **locate where a behavior comes from** inside the model with causal interventions, and **post-train models to correct errors** (SFT, RL with verifiable rewards, low-rank updates) while measuring side effects such as losing previously correct answers. I also bring causal discovery experience on single-cell perturbation data (LG AI Research; NeurIPS 2026 AI4DD workshop) and a neuroscience background. Seeking to build and evaluate LLM systems that reason faithfully over clinical protocols and regulatory logic.

EDUCATION

Ph.D. in Machine Learning, Georgia Institute of Technology

Anticipated 12/2026

B.S. in Neuroscience (Faculty Honors), Georgia Institute of Technology

RESEARCH EXPERIENCE

Graduate Research Assistant (PI: C. S. Mitchell, Ph.D.)

08/2021–present

Laboratory for Pathology Dynamics, Georgia Tech & Emory University School of Medicine

- **[Goal-Aware LLM Alignment under Evidence and Eligibility Constraints]**
 - Ongoing research project | 2026–present
 - Design **counterfactual evaluations** to test whether LLMs update eligibility judgments when external rules change, yet keep them invariant when only the patient's stated goal changes, in controlled synthetic treatment-decision tasks built on attribute ranges from a published patient-preference survey.
 - Localize the model's rule-application failure with **activation patching**: identified a small set of attention heads (2% of heads) whose activations, transplanted from a corrected reference model, recover the reference's eligibility correction on items held out from selection, whereas depth-matched random heads do not.
 - Develop **component-restricted post-training** (supervised fact loss + reinforcement learning with verifiable rewards, with low-rank updates confined to the selected heads) and evaluate it against budget-matched random-head and full-parameter baselines on the trade-off between rule correction and side effects (wrongly excluding permitted options, losing previously correct answers).
- **[Telling Apart What Fine-Tuning Against Wrong Diagnostic Suggestions Gains and Loses]**
 - Showed that KL-regularized LoRA fine-tuning of Llama-3.1-8B-Instruct to resist wrong diagnostic suggestions (also with GRPO, and repeated on Qwen2.5-7B-Instruct) raises accuracy under wrong suggestions (30.5% → 71.5%) but lowers it under correct ones (89.1% → 62.2%), and that what it loses is **the bias toward following suggestions, not discrimination**: telling correct from wrong suggestions improves (AUROC 0.684 → 0.741).
 - Separated the two changes with **cross-model activation patching** at layer 11: the bias change is carried at the suggestion tokens and the discrimination change at the answer-option tokens, where a low-dimensional subspace carries the effect of the key clinical findings.
 - **Steering only the suggestion tokens** with one mean activation difference recovers 59% of the lost accuracy under correct suggestions while keeping 79% of the discrimination gain (vs. 39–44% for weight or activation interpolation), holding on 951 held-out pairs; a constant logit adjustment matches its accuracy, and a probe-gated variant directs steering toward inputs with missing findings.
 - Under review at ACL Rolling Review (2026).
- **[Seven Medical Reasoning Biases, One Causal Direction]**
 - Investigated whether answer-suggesting cues in medical QA correspond to separate internal mechanisms or share an **activation-space control point**, using contrastive prompts and causal steering across seven cue types and three LLM families.
 - Found that an opinion-contrast direction reduced selection of cue-suggested wrong answers across cue types, while improvements in answer accuracy were smaller. Compared shared, cue-specific, and random directions to **separate detectable representations from effective interventions**.
 - Under review at ACL Rolling Review (2026).

- **[Recovery Is Not Provenance: A Source-Substitution Audit of VLM Activation Patching in Medical VQA]**
 - Exposed a **source-attribution gap in VLM causal tracing**: activation patching that repairs an answer flipped by an authoritative incorrect text prompt need not restore evidence from the source image. Introduced source substitution, a paired identification test that swaps the donor case while fixing the recipient and patch site.
 - Across 5 VLMs on SLAKE with heads localized on VQA-RAD, **different-case donors reproduced 98–103% of same-item recovery** with no detected same-item advantage, robust across decoder depth, 99 alternative donor assignments per model, and real-image recipient substitution.
 - Validated the intervention itself: forward and reverse patches move margins oppositely, identity replay is exact, and selected heads exceed all 95 matched-random component sets by $\geq 3.5\times$.
 - *Under review at AAAI 2027.*

ADDITIONAL RESEARCH EXPERIENCE

Research Intern, LG AI Research (Bio Intelligence Lab)

05/2026–08/2026

- Developed **DCCD-J**, a differentiable method for recovering causal graphs from interventional data under latent confounding and feedback loops. Diagnosed why the SOTA baseline's alternating covariance estimation collapses on real data and replaced it with a jointly optimized Cholesky parameterization that is positive-definite by construction while preserving the score-based identifiability guarantee; best held-out likelihood on all 5 benchmarks, confounder-recovery F1 0.93 vs. 0.23 under dense confounding. Co-first author; **NeurIPS 2026 Workshop on AI for Drug Discovery (AI4DD)**.
- Replaced standard expression-based variable selection with **attribution from single-cell foundation models** (scFoundation, scLong) to choose the gene set for DCCD-CONF regulatory inference.

Graduate Research Assistant, Georgia Tech

08/2021–present

- **HiCo-EL**: adapted **hierarchical contrastive learning to bi-encoder entity linking**, organizing each encoder space by the ontology type hierarchy (UMLS semantic types; CTD MEDIC) with every concept as a leaf, while an alignment loss keeps the two spaces compatible for retrieval. Best R@1 on MedMentions (+3.7% over ArboEL) and NCBI Disease (+1.8%), with **+9.6% R@1 on diseases unseen in training**. Supervised two junior researchers. **EMNLP 2026**.
- **Clinical trial NLP**: PICO information extraction from clinical trial literature for meta-analysis and drug repurposing (TrialSieve, Bioengineering 2025); relevance prediction, extractive summarization, and NER for screening clinical cohort studies (CCS Explorer, IEEE BigData 2022 Oral).
- Interpretable ML for cognitive assessment: characterized how education, sex, and age shape Rey Complex Figure performance in cognitively normal adults (Journal of Clinical Medicine, 2025).

Computational Neuroscience Rotating Fellow (PI: C. Rozell, Ph.D.)

01/2021–05/2021

Department of Electrical Engineering, Georgia Tech

- Applied embedding-based similarity models and probabilistic latent-variable methods to characterize clinical depression phenotypes.

Undergraduate Research Fellow and ALS Team Lead (PI: C. S. Mitchell, Ph.D.)

12/2016–05/2020

Laboratory for Pathology Dynamics, Georgia Tech & Emory University School of Medicine

- Led multi-year ML-driven meta-analysis integrating heterogeneous preclinical datasets (>2,000 extracted data points); built multi-scale models combining unsupervised learning, global optimization, and dynamic system analysis.
- Led an NIH grant and a team of 5 researchers, resulting in 9 publications.

SKILLS

- **LLM training & customization**: Supervised fine-tuning, KL-regularized LoRA, reinforcement learning with verifiable rewards (GRPO), component-restricted updates, domain adaptation to clinical decision tasks, robustness fine-tuning.
- **Evaluation & failure analysis**: Counterfactual evaluation and benchmark design for high-stakes clinical tasks, faithfulness and source-provenance auditing, side-effect measurement (over-exclusion, regression on correct answers), activation patching & causal tracing, activation steering, linear probing.
- **Causal & graph modeling**: Causal discovery from interventional data under latent confounding and feedback loops; single-cell foundation model attribution.
- **Engineering**: Python (PyTorch, HuggingFace, Accelerate, DeepSpeed), multi-GPU training and inference, W&B; MATLAB, R.
- **Languages**: Native in Korean; fluent in English.

PUBLICATIONS

Under review:

- [1] **Lee, A. J. B.**, & Mitchell, C. S. Recovery Is Not Provenance: A Source-Substitution Audit of VLM Activation Patching in Medical VQA. Under review at AAAI 2027.
- [2] **Lee, A. J. B.**, & Mitchell, C. S. Seven Medical Reasoning Biases, One Causal Direction. Under review at ACL Rolling Review (2026).
- [3] **Lee, A. J. B.**, & Mitchell, C. S. Telling Apart What Fine-Tuning Against Wrong Diagnostic Suggestions Gains and Loses. Under review at ACL Rolling Review (2026).

Peer-reviewed conference papers:

- [4] **Lee, A. J. B.**, Bathala, P., Malhotra, P., Ye, C., & Mitchell, C. S. (2026). HiCo-EL: Encoding Ontology Structure into Bi-Encoder Entity Linking. EMNLP 2026.
- [5] Al-Hussaini, I., An, D. N., **Lee, A. J.**, Bi, S., & Mitchell, C. S. (2022). CCS Explorer: Relevance Prediction, Extractive Summarization, and Named Entity Recognition from Clinical Cohort Studies. IEEE International Conference on Big Data (IEEE BigData 2022), Oral.

Peer-reviewed workshop papers:

- [6] **Lee, A. J. B.***, Kang, J.*, Kim, S.*, & Jeong, H. (2026). Separating Co-expression from Regulation in Cyclic Causal Discovery with Latent Confounders. NeurIPS 2026 Workshop on AI for Drug Discovery (AI4DD). *Equal contribution.

Peer-reviewed journal articles:

- [7] He, X., Camacho, D., Moukheiber, L., Iyer, M., Zhao, B., Ye, C., Nursal, B., Guo, X., **Lee, A. J. B.**, & Mitchell, C. S. (2026). SemNet Explorer: An Evidence-Grounded Knowledge Graph-LLM Framework for Multi-Scale Mechanistic Reporting Across Biomedical Domains. Big Data and Cognitive Computing, 10(6), 171.
- [8] Fay, M. E., Tandon, R., Latham, T., **Lee, A. J.**, Rankine-Mullins, A. E., Reid, M., Mitchell, C. S., Ware, R. E., & Lam, W. A. (2026). Artificial Intelligence-Based Analysis of Central Nervous System Vasculopathy in Pediatric Sickle Cell Anemia. American Journal of Hematology, 101(5), 998–1004.
- [9] **Lee, A. J. B.**, Zhao, B., Lah, J. J., John, S. E., Loring, D. W., & Mitchell, C. S. (2025). Education, Sex, and Age Shape Rey Complex Figure Performance in Cognitively Normal Adults: An Interpretable Machine Learning Study. Journal of Clinical Medicine, 14(21), 7562.
- [10] Kartchner, D., Turner, H., Ye, C., Al-Hussaini, I., Nursal, B., **Lee, A. J. B.**, Deng, J., Curtis, C., Cho, H., Duvaris, E. L., Jackson, C., Shanks, C. E., Tan, S. Y., Ramalingam, S., & Mitchell, C. S. (2025). TrialSieve: A Comprehensive Biomedical Information Extraction Framework for PICO, Meta-Analysis, and Drug Repurposing. Bioengineering, 12(5), 486.
- [11] **Lee, A. J. B.**, Bi, S., Ridgeway, E., Al-Hussaini, I., Deshpande, S., Krueger, A., Khatri, A., Tsui, D., Deng, J., & Mitchell, C. S. (2025). Restoring Homeostasis: Treating Amyotrophic Lateral Sclerosis by Resolving Dynamic Regulatory Instability. International Journal of Molecular Sciences, 26(3), 872.
- [12] **Lee, A. J. B.**, Kittel, T. E., Kim, R. B., Bach, T. N., Zhang, T., & Mitchell, C. S. (2022). Comparing Therapeutic Modulators of the SOD1 G93A Amyotrophic Lateral Sclerosis Mouse Pathophysiology. Frontiers in Neuroscience, 16, 1111763.

Conference Proceedings with Presentation:

- [1] Deng, J., Bi, S., Tsui, D., **Lee, A.**, Al-Hussaini, I., & Mitchell, C. S. (2022). Homeostatic re-stabilization of SOD1-G93A ALS via combination therapy with repurposed drugs. International Motoneuron Society, Banff, Canada.
- [2] Krueger, A., Bi, S., **Lee, A.**, & Mitchell, C. S. (2021). In silico multi-factorial, multi-scalar models of system stability for ALS combination therapies. AAPM Conference on Data Science in Medical Physics.
- [3] Bi, S., Krueger, A., **Lee, A.**, & Mitchell, C. S. (2021). In-silico multi-factorial system stability to identify promising SOD1-G93A ALS mouse combination treatments. Society for Neuroscience.
- [4] Bi, S., Krueger, A., Ridgeway, E., Khatri, A., Deshpande, S., **Lee, A.**, & Mitchell, C. S. (2021). Dynamic Network Stability Analysis for Prioritizing Experimental Combination Therapies for ALS. American Neurological Association, 146th Annual Meeting.
- [5] **Lee, A.**, Lee, B., Ridgeway, E., & Mitchell, C. S. (2019). Machine learning enhanced dynamic meta-analysis for preclinical ALS combination therapy optimization. Society for Neuroscience, Chicago, IL.
- [6] **Lee, A.**, Lee, B., Ridgeway, E., & Mitchell, C. S. (2019). Dynamic Meta-Analysis to Assess Multi-factorial Neuropathology. American Neurological Association, St. Louis, MO.
- [7] **Lee, A.**, Lee, B., Ridgeway, E., & Mitchell, C. S. (2019). Machine learning enhanced dynamic meta-analysis for multi-factorial disease prediction. MLSE Conference, Atlanta, GA.
- [8] Deshpande, S., Lee, B., **Lee, A.**, Ridgeway, E., & Mitchell, C. S. (2019). Dynamic Meta-Analysis to Assess Multi-factorial Disease. SEMDA Medtech Conference, Alpharetta, GA.
- [9] **Lee, A.**, & Mitchell, C. S. (2019). Discovering potential combinational treatments of ALS using a computational model of G93A mouse. Georgia Tech Undergraduate Research Spring Symposium, Atlanta, GA.

- [10] **Lee, A.**, Travaglino, S., & Mitchell, C. S. (2018). Dynamic Meta-Analysis Approach to In Silico Models of Temporal Multi-scalar Neurophysiology. Biomedical Engineering Society, Atlanta, GA.
- [11] **Lee, A.**, Travaglino, S., & Mitchell, C. S. (2017). Towards syncing polytherapy to the multi-factorial dynamics of the SOD1 G93A ALS pathology. Society for Neuroscience, Washington, D.C.
- [12] Kittel, T., **Lee, A.**, Nguyen-Bach, T., & Mitchell, C. S. (2017). Data enabled design of multi-factorial disease treatment paradigms: a case study of ALS. National Academy of Engineering Regional Meeting, Atlanta, GA.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Georgia Tech, Dept. of Biomedical Engineering

Summer 2025

- Graded homework, held weekly office hours, and reviewed concepts for BMED2400 (Bioengineering Statistics).
- Nominated for departmental award recognizing outstanding teaching and mentorship.
- Supervised two junior researchers on HiCo-EL through to an EMNLP 2026 acceptance.

HONORS / FELLOWSHIPS

BME Department Outstanding Teaching & Mentorship Award — Nominee

2025

National Science Foundation Graduate Research Fellowship — Honorable Mention

05/2020

NIH T32 Training Fellowship in Computational Neuroengineering

05/2020

President's Undergraduate Research Project Award

President's Undergraduate Research Travel Award

10/2017

Zell Miller Scholarship (full-tuition merit-based scholarship)